Research Statement

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Rapidly evolving hardware technologies and applications continually create new gaps between what hardware can provide and what applications require. As a systems researcher, my work addresses a fundamental challenge: how to adapt system software to bridge the gaps between evolving hardware capabilities and application demands.

Emerging hardware technologies, from high-density SSDs to CXL memory and specialized accelerators, continue to reshape computing infrastructure. Meanwhile, applications, from cloud workloads to AI agents, introduce new demands for performance and scalability. However, existing system software stacks are often designed for earlier generations of hardware and applications, preventing them from fully exploiting new hardware capabilities or efficiently supporting emerging workloads. My research bridges these gaps by rethinking storage abstractions, execution architectures, and memory management.

Looking ahead, the accelerating evolution of hardware and applications presents growing challenges for future system development and optimization. My long-term vision is to build evolvable computing infrastructure in which system software can automatically and continuously co-evolve with emerging hardware technologies and application demands.

Evolving System Software Across Storage, I/O, and Memory for the Cloud

My current research focuses on evolving system software to unlock emerging hardware capabilities and meet the performance and efficiency demands of large-scale cloud workloads. My work explores this challenge across storage abstractions, I/O processing, and memory management.

Storage Abstractions for Emerging Hardware. High-density SSDs [1] offer greater capacity at lower cost than previous SSDs, but suffer from limited endurance and performance due to the inherent limitations of high-density flash and increased write amplification caused by coarser-grained indirection units. To address these challenges, I designed CSAL [2], a new storage abstraction that enables high-density SSDs to achieve both high performance and endurance. CSAL moves firmware-level indexing into host software and integrates it with tiering metadata, creating a unified indexing structure that supports finer-grained indexing while controlling memory footprint through page-table and swapping techniques. In addition, CSAL adopts log-structured data management across fast and slow tiers, transforming small random writes into flash-friendly writes on high-density SSDs and managing flash garbage collection in software. CSAL enabled the deployment of high-density SSDs in large-scale cloud data centers [3] and has since been adopted by multiple production storage systems [4, 5, 6, 7, 8].

I/O Processing for Efficiency. As storage hardware performance continues to improve rapidly, efficient I/O processing becomes increasingly important for large-scale cloud infrastructure. Modern storage servers often rely on dedicated polling CPU cores to maintain low latency, resulting in substantial energy waste when I/O demand is low. To address this challenge, I designed Sandman [9], a new CPU scheduler that dynamically puts CPU cores to sleep based on workload demands to save energy. To maintain low latency, Sandman leverages hardware cache coherence to efficiently wake sleeping CPU cores, avoiding both software and hardware wakeup overhead. It also monitors incoming I/O requests from network queues to detect bursts at microsecond timescales, enabling fine-grained CPU scheduling without sacrificing responsiveness.

On cloud compute nodes, however, I/O processing faces a more fundamental challenge. Unlike storage nodes, these nodes must reserve most CPU resources for customer workloads, leaving limited CPU capacity for infrastructure software stacks. As storage hardware becomes faster, this limited CPU capacity increasingly constrains the maximum I/O performance these stacks can deliver. To address this bottleneck, I designed Ristretto [10], a new storage architecture that combines ASIC acceleration with programmable ARM SoCs to achieve high performance with substantially fewer CPU resources. Ristretto offloads compute-intensive but relatively stable functions to ASICs. Meanwhile, it retains cloud-specific functionality on ARM SoCs to preserve software flexibility. This architecture has been deployed in large-scale cloud data centers, and the work received the FAST'26 Best Paper Award.

Memory Management Beyond Traditional Boundaries. Memory is a critical resource in cloud data centers, yet traditional memory management focuses primarily on capacity rather than other resource dimensions and remains largely confined to individual servers. To overcome the limitations of capacity-centric memory management, I designed RamRyder [11], a new memory system that enables mostly independent allocation of memory capacity and bandwidth with robust performance isolation. RamRyder shifts memory channel management from hardware to software through a new OS-level channel abstraction and page-to-channel allocation policies across DRAM and CXL memory tiers. This work establishes a new direction in software-defined, open-channel memory systems, demonstrating performance and utilization benefits using realistic cloud workload traces.

To overcome the limitations of single-node memory management, my ongoing work, Aspen [12], explores a new memory abstraction for efficient memory sharing across cloud virtual machines (VMs). Existing memory deduplication mechanisms are largely confined to individual servers, limiting their ability to eliminate redundant memory copies across nodes. Aspen leverages emerging CXL fabrics [13] to enable VMs on different nodes to share physical memory pages through a shared memory pool. It decouples VM memory management from physical memory placement and explores lightweight memory mapping and reference-based sharing mechanisms to reduce memory duplication and data movement across cloud data centers.

Future Directions: From System Adaptation to Co-Evolution

The accelerating evolution of hardware and applications is outpacing traditional approaches to system adaptation. My future research will first focus on rethinking system designs to meet emerging application demands, particularly those introduced by AI agents. I will further explore how memory systems can evolve beyond node- and rack-level boundaries toward datacenter-scale computing. My ultimate goal is to achieve automatic and continuous co-evolution of system software with emerging hardware technologies and application demands.

Rethinking Storage Architectures, Interfaces, and I/O Systems for AI Agents. Today's storage systems are largely designed for traditional cloud workloads, e.g., databases, big data analytics, and serverless computing. However, the rise of AI agents introduces new challenges for storage systems, as agents can dynamically create and terminate execution environments across cloud and edge, generate large amounts of intermediate and persistent context data, and concurrently access large shared datasets. These emerging demands not only pose new challenges to performance but also require new agent-native storage systems.

To address these challenges, I will investigate new storage architectures, interfaces, and I/O systems for AI agents. First, I will investigate new architectures that can efficiently manage large

volumes of context data across distributed agents. Second, I will design new interfaces that enable secure data sharing and concurrent access among collaborating agents across cloud and edge. Third, I will develop new I/O systems that schedule I/O requests based on the characteristics of agentic workloads to achieve low latency and high resource utilization.

Datacenter-Scale Unified Memory for Heterogeneous Computing. Scaling AI infrastructure requires heterogeneous computing resources (e.g., CPUs, GPUs, and TPUs), yet faces two fundamental memory challenges. First, computing is increasingly constrained by the memory capacity of a single node or even an entire rack. Second, efficient data sharing across heterogeneous computing resources remains challenging due to fragmented memory spaces and costly data movement. These challenges are becoming increasingly severe as AI workloads continue to scale and computing hardware becomes more heterogeneous.

To overcome this limitation, I will investigate unified memory systems that can scale across entire datacenters to support heterogeneous computing resources. First, I will explore how to efficiently interconnect memory resources by leveraging emerging scale-up (e.g., NVLink and CXL) and scale-out (e.g., Ethernet/InfiniBand RDMA) networking technologies. Second, I will investigate memory management across the memory hierarchy (e.g., HBM, DRAM, and flash) and distributed nodes, enabling transparent data movement across multiple tiers and nodes to meet the capacity, bandwidth, and latency requirements of diverse agentic workloads. Third, I will explore new protocols and algorithms that can provide scalable memory coherence across distributed nodes to enable efficient data sharing throughout the datacenter.

Towards Autonomous Co-Evolution of System Software. System software must continuously evolve to support emerging hardware and applications. However, developing new systems or adding features to existing ones requires substantial human effort and long development cycles. Today's AI agents have demonstrated strong capabilities in coding and research across various domains, but system software remains highly complex, where even small implementation errors can have severe consequences at the datacenter scale.

My goal is to harness AI agents to accelerate system development for emerging hardware and enable automatic optimization based on workload characteristics. To achieve this goal, I will investigate three directions. First, I will explore how to raise the abstraction level of system implementation by decoupling core system components from specialized features, allowing AI agents to operate at a higher level without compromising system stability. Second, I will investigate the use of system specifications as "prompts" to help AI agents understand system requirements and solve complex systems problems. Third, I will develop new verification and testing techniques to ensure the correctness and robustness of agent-generated system code.

Summary

My prior research in evolving system software for cloud infrastructure provides a strong foundation for my long-term vision. I aim to push the frontier of systems research while developing practical technologies with real-world impact. I look forward to contributing to a collaborative and diverse research environment, working with students and researchers, and helping shape the next generation of computing systems.

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In Submission.

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